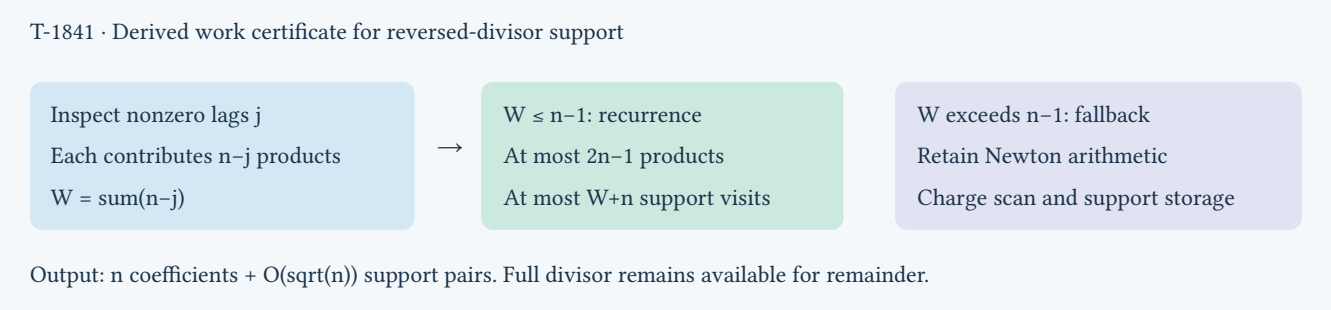
Linear quotient work from weighted support

T-1841 · 2026-11-07 · Native prime-field polynomial kernel

The quotient dispatcher now accepts several subleading divisor terms when their total contribution count fits a linear work certificate. It strictly contains the preceding one-term class. This broadens the usable structure without assuming that every sparse polynomial is cheap or that the source workload frequently has this structure.



Derived identity. With reversed dividend A and divisor B, leading alpha nonzero and relevant nonzero lags J, coefficient division gives $C_i = (A_i - \text{sum of } B_j\, C(i-j))/\text{alpha}$. Induction establishes B C = A modulo X^n. Reversing C returns the exact quotient; lower original divisor coefficients still enter the remainder.

Work certificate. Accept W = sum over J of (n−j) ≤ n−1. Skip zero products and monic scaling. Exact field products are u Z(C) + sum of $Z_j(C)$, at most u n + W ≤ 2n−1, with u inversions; u indicates a nonmonic divisor. Traverse sorted support only through the first future lag: exactly W productive visits, at most W+n total visits. Scan and output reversal are linear. Support size s satisfies s(s+1)/2 ≤ n−1, giving O(sqrt(n)) pairs plus the n-entry output. Input, remainder and recursive storage remain additional.

Deterministic software check	Cases	Result
Accepted weighted bands	2,080	Exact products, visits and quotient
Multiple-term accepted bands	896	Up to 15 subleading terms
Exact work-cap boundary	640	Accepted at W = n−1
Rejected bands	4,192	Same Newton field counts
Inherited source checks	1,108 points	24 windows; all pass
Additional recoveries	617	5,929 calls; zero unresolved
Inherited public comparisons	63	511 calls; all pass

Scope and controls. Exhaustive support masks for lengths up to eight accompany named and longer controls. Counts include deliberate overlap. Single-term, dense, short-quotient, remainder, matrix, resultant and singleton controls remain passing. Normal build has seven input pins and excludes audits. This is software validation, not elapsed-time evidence. Original 4,563-case cohort unchanged; no cloud or full Cargo rerun. PCC-20261107-98493f is unsigned producer evidence.

Sources and next step. Standard coefficient division recurrence: FLINT nmod_poly series-division documentation; exact predecessor source. KB unavailable. DERIVATION.md proves the bound and records assumptions. Next trace support lengths and all source-phase costs in actual public-point recovery before claiming an aggregate reduction. Historical evidence and canonical research graphs are preserved.